

OMIS 114 Final Presentation
December 2025

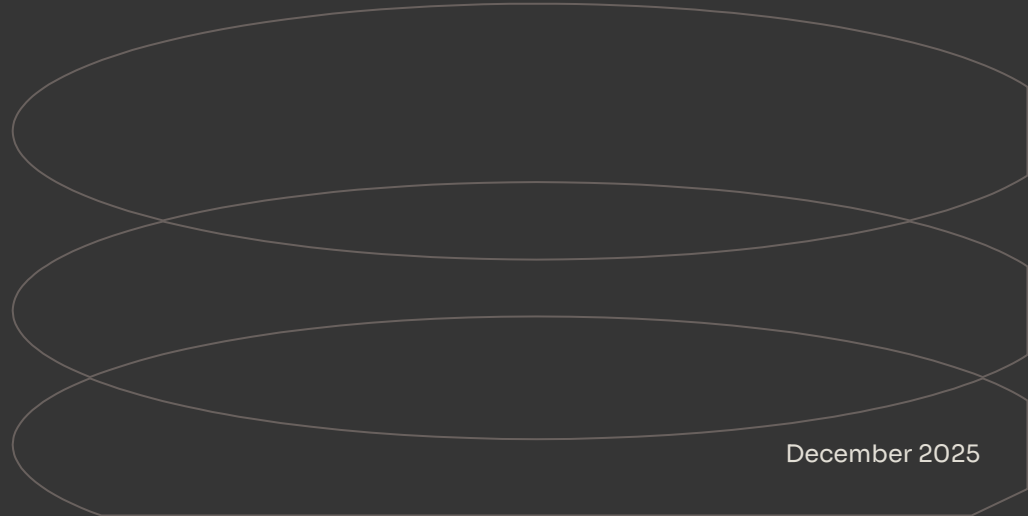
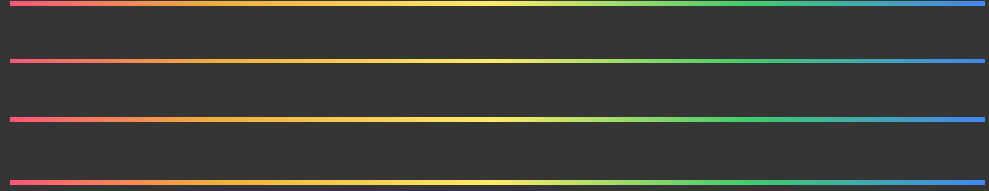
Sales Data Analysis & Customer Insights

By Jeevan Brar, Dominic Paine, and Tristan Poul

Agenda

1.
Data Overview & Preparation
2.
Exploratory Analysis & Key Findings
3.
Predictive Modeling
4.
Behavioral Segmentation
5.
Recommendations & Conclusion

Data Overview & Preparation



Data Quality & Initial Exploration

1

Data dimensions:

Rows: 10,000

Columns: 15

2

Data Types:

float64, string, datetime,
int64

3

Data Quality

Examples

	customer_id	age	gender	location	income	registration_date	purchase_date	product_category	brand	price	quantity	revenue	purchase_frequency	avg_order_value	customer_lifespan
0	CUST_00001	40.0	<NA>	<NA>	33889.0	2025-04-25	2025-07-09	Home & Garden	Brand H	58.75	2	117.51	3	140.64	0.41
1	CUST_00002	30.0	Male	San Francisco	20000.0	2025-02-16	2025-08-25	Automotive	Brand F	177.15	2	354.31	2	369.55	0.60
2	CUST_00003	NaN	Male	Boston	58321.0	2025-05-25	2025-09-13	Sports	Brand G	133.91	2	267.81	4	275.07	0.33
3	CUST_00004	39.0	Male	<NA>	39590.0	2022-11-12	2023-09-04	Books	Brand H	16.46	6	98.75	2	79.62	2.86
4	CUST_00005	42.0	Female	Phoenix	23078.0	2025-01-06	2025-02-16	Sports	Brand G	94.95	1	94.95	5	97.28	0.71

age: float64, **gender:** string, **registration_date:** datetime, **quantity:** int64

	Missing Values	Missing %	Dtype	Unique
age	792	7.92	float64	59
income	557	5.57	float64	7753
gender	524	5.24	string	3
location	408	4.08	string	20

Building a Clean Foundation for Analysis

Handled Missing Values

Imputed missing 'Income' and 'Purchase Frequency' using the median to avoid skewing the data.

Corrected Data Types

Converted 'Date' column to datetime objects for time series analysis.

Removed Duplicates

Dropped 55 duplicate rows to ensure data integrity.

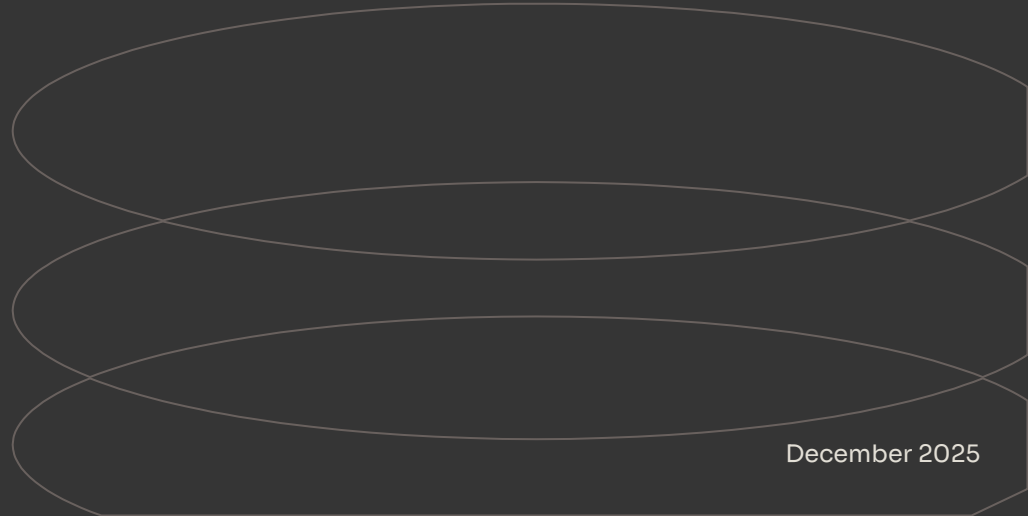
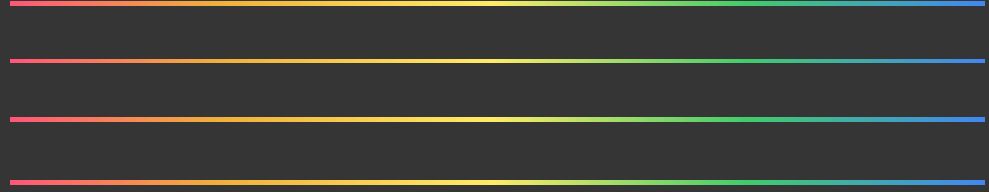
Feature Engineering

Created 'Customer Lifetime Value' (CLV) and 'Purchase Recency' to provide deeper business insights.

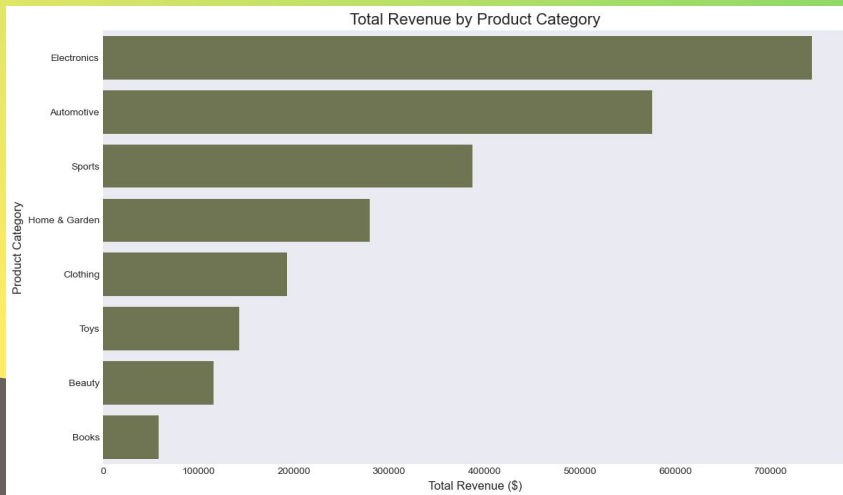
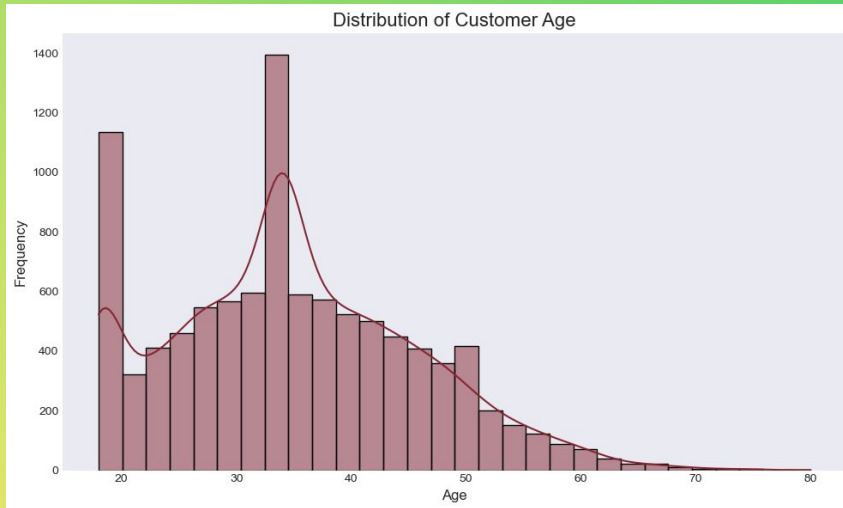
Encoded Categorical Variables

Used one-hot encoding for variables like 'Location' and 'Gender' to prepare them for machine learning.

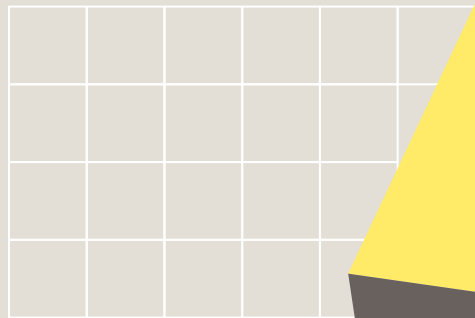
Exploratory Analysis & Key Findings



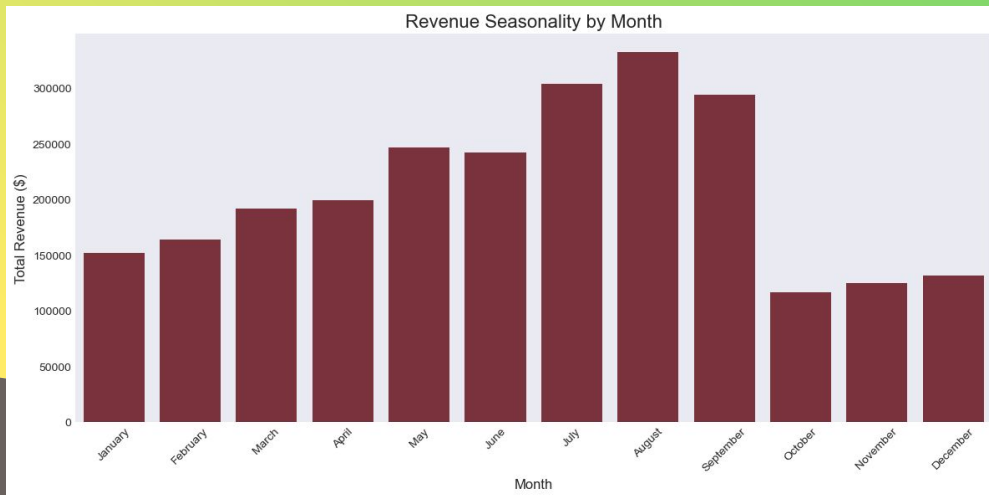
Who Are Our Customers and What Are They Buying?



Analyzing Revenue Trends & Order Values

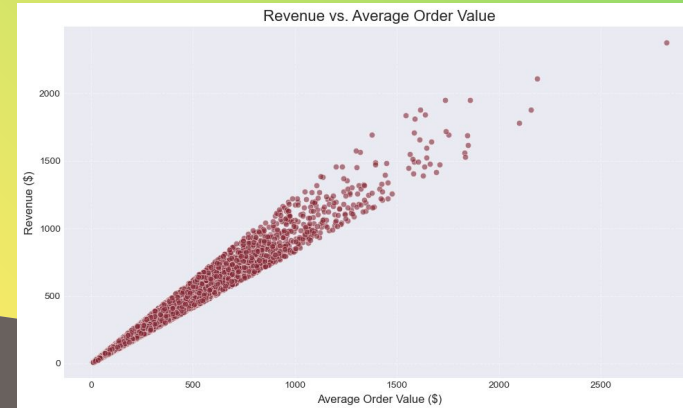
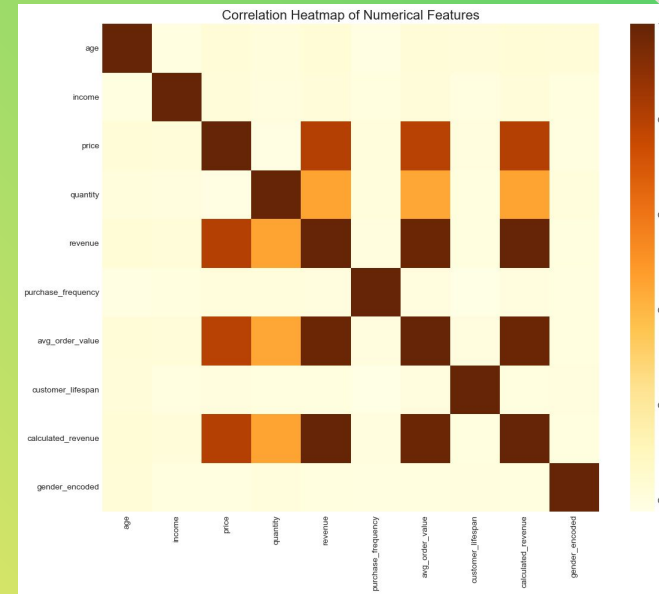


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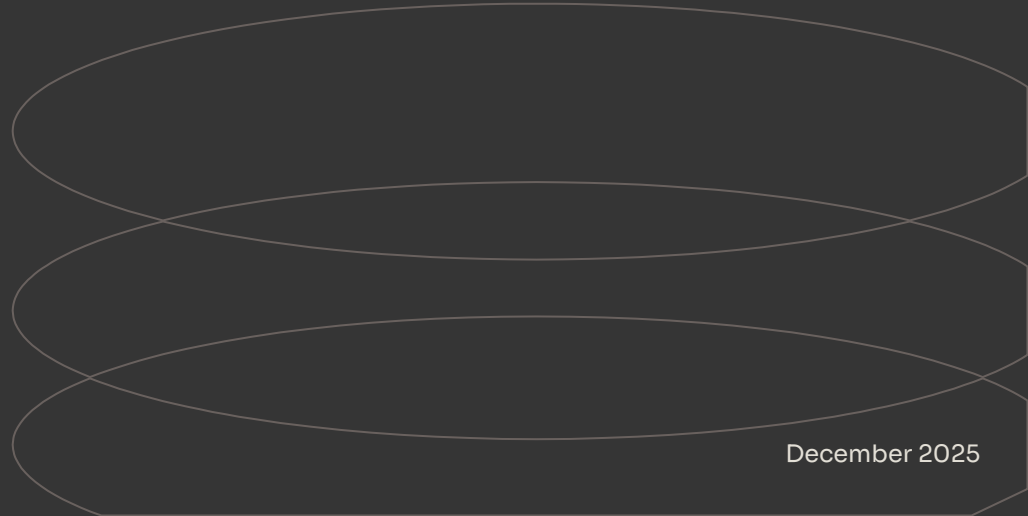
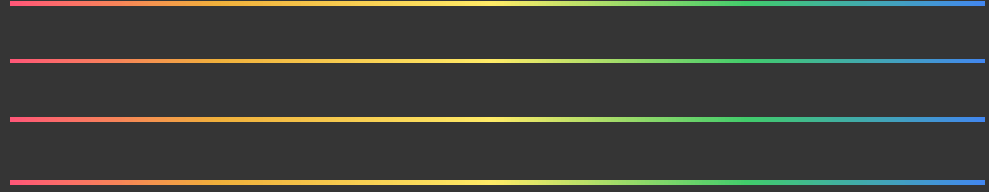


Correlations and Customer Behavior

	Feature_1	Feature_2	Correlation
46	avg_order_value	calculated_revenue	0.986810
35	revenue	avg_order_value	0.986810
20	price	revenue	0.801652
24	price	calculated_revenue	0.801652
22	price	avg_order_value	0.792096



Predictive Modeling



Two Core Questions

Model 1: Classification

Which customers are high value?

Target:
High_Value_Customer (top 30% by revenue)

Model: Random Forest Classifier

Use Case: Prioritize customers for retention.

Model 2: Regression

How much revenue will this customer/order generate?

Target: revenue

Model: Gradient Boosting Regressor

Use Case: Forecast revenue; support budgeting and planning

How They Connect

First model explains **who** the high value customers are and the second model explains **how much** they're spending.

Shared Inputs: age, income, gender, purchase_frequency, avg_order_value, customer_lifespan, location, product_category, brand

Model 1: High-Value Customer Classification

How did the Classification model perform?

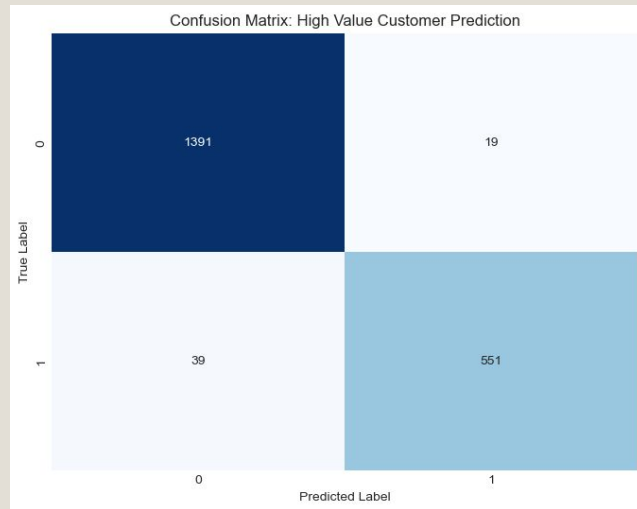
Random Forest Model

Accuracy: 0.9710

Precision: 0.97

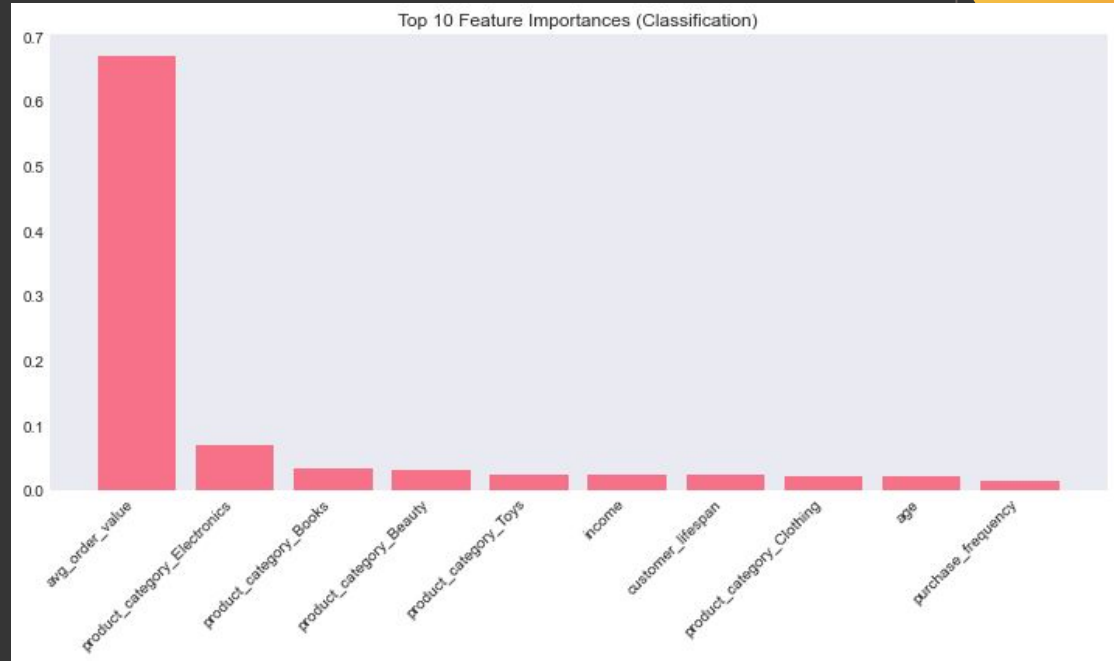
Recall: 0.97

The **Precision** tells us that our positive predictions are reliable, and **Recall** shows that the model accurately identifies true high-value customers.



Key Predictors of High-Value Behavior

- AOV is the biggest driver of high-value behavior
- Many intuitive features lack importance: income, age
- Product categories are the next biggest feature importances



Model 2: Revenue Prediction

Model 2: Revenue Prediction

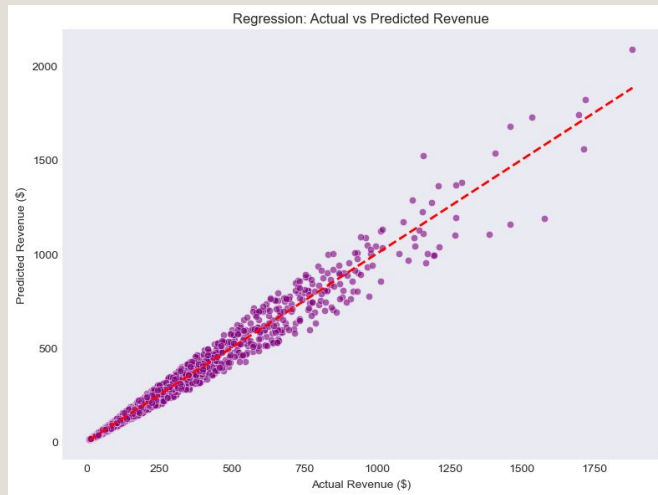
Gradient Boosting Regressor Model

Mean Absolute Error (MAE): \$25.27

Root Mean Squared Error (RMSE): \$41.78

R²: 0.9716

The model captures a **large share of the variation in customer revenue**. It's not accurate enough to predict exact dollar values, but it could be used to **rank customers by expected value**.



From Models to Business Action

Classification → Who to Target

Key tool for identifying high-value customers early.

Goal: Find customers who spend in upper 30%.

Prioritize the identified people for VIP programs, retention, and promotions.

Regression → How Much They're Worth

Great tool for forecasting revenue contributions.

Goal: Make revenue forecasts more accurate

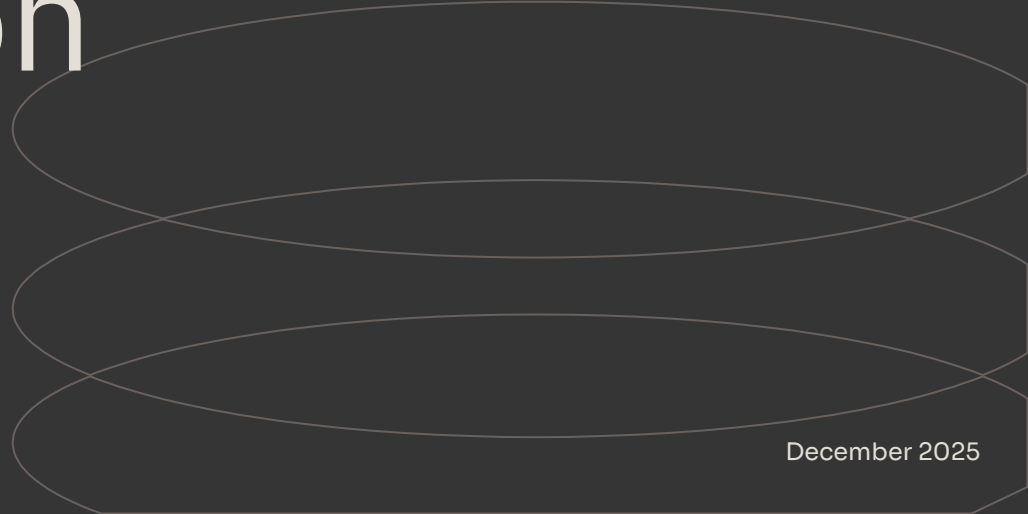
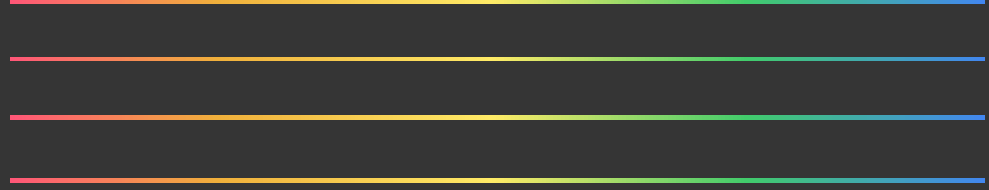
This can support budgeting & inventory planning

Interpretation

These models work together to tell us **who will likely be high value and how much they might contribute.**

This leads naturally into a more robust segmentation: **RFM.**

Behavioral Segmentation



Beyond Prediction: RFM Segmentation

- Recency: Days since last purchase
- Frequency: Number of purchases
- Monetary: Total revenue
- Each scored 1–4 → 7 customer segments
- Creates actionable behavioral segments

Which Segments Drive Revenue?

- **Loyal Customers** and **Big Spenders** drive the largest share of reliable revenue.
- **At Risk** customers represent the largest pool of customers, which is an action point for the future.
- **Champions** deliver exceptional value per customer.

CUSTOMER INSIGHT ANALYSIS (RFM)

[Analysis] Customer Segments Created:

Segment

At Risk 0.2486

Loyal Customers 0.2356

Lost 0.1264

Promising New 0.1254

Big Spenders 0.1246

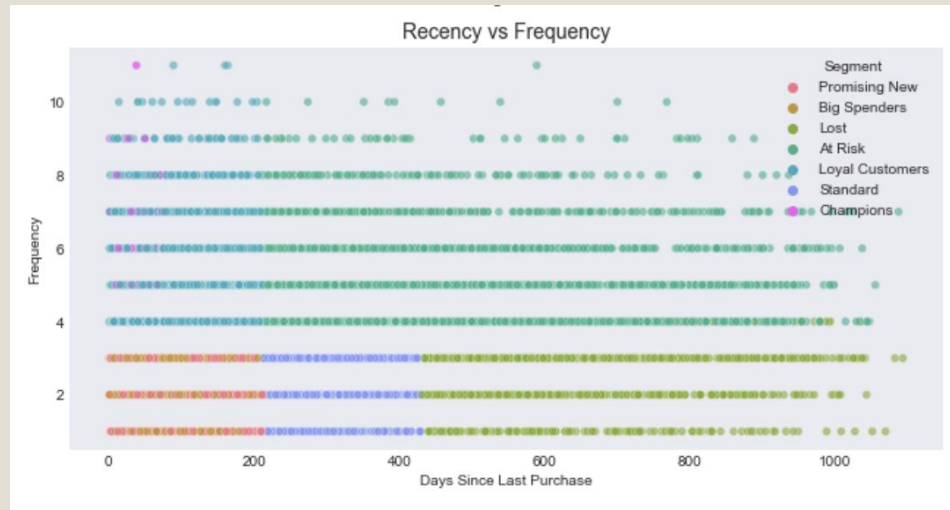
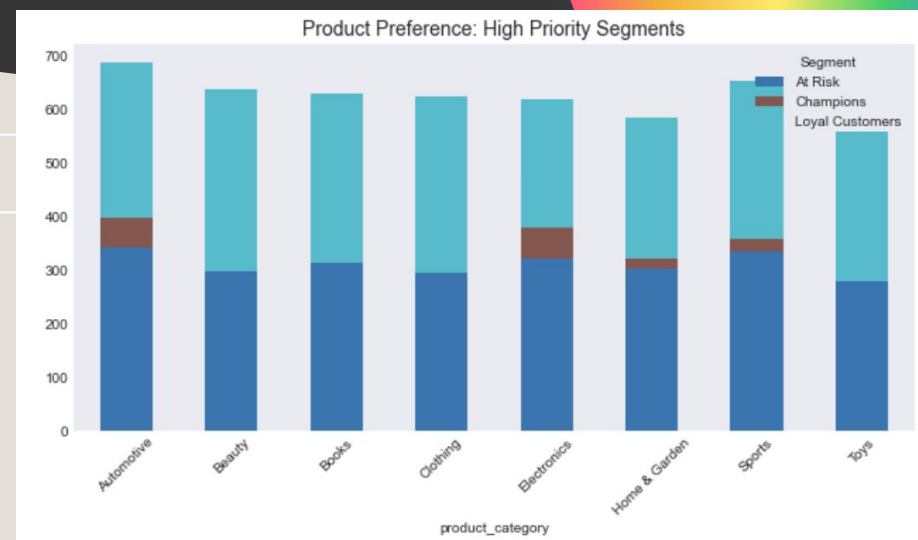
Standard 0.1236

Champions 0.0158

Name: proportion, dtype: float64

Behavioral View of Customer Segments

- Champions consistently buy often
- At Risk buyers used to buy often but haven't recently
- Segments differ in product preference



Recommendations & Conclusion

Key Customer Value Metrics

Turning segmentation into quantitative business insights.

1

Total Revenue Analyzed:

\$2.49M

2

Champions:

158

3

Average CLV:

\$249.74

4

At Risk Customers

2486

Data-Driven Strategy Recommendations

1. Win-Back At Risk Customers

- Personalized discounts
- Reminder campaigns
- High ROI based on past spending

2. VIP Program for Champions

- Early access to high-preference categories
- Exclusive loyalty perks
- Increase retention & AOV

3. Smarter Acquisition Targeting

- Build lookalikes from Champion profiles
- Optimize ad spend

Implementation Plan

1

Automate RFM + model scoring weekly

2

Push segment and score labels into CRM

3

Trigger targeted campaigns per segment

4

Monitor KPIs: retention, AOV, CLV